

Industrial Internship Report on
" Prediction of Agricultural Crop Production in India"

&

" Crop and Weed Detection using YOLO"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My projects were "Prediction of Agricultural Crop Production in India" and "Crop and Weed Detection using YOLO".

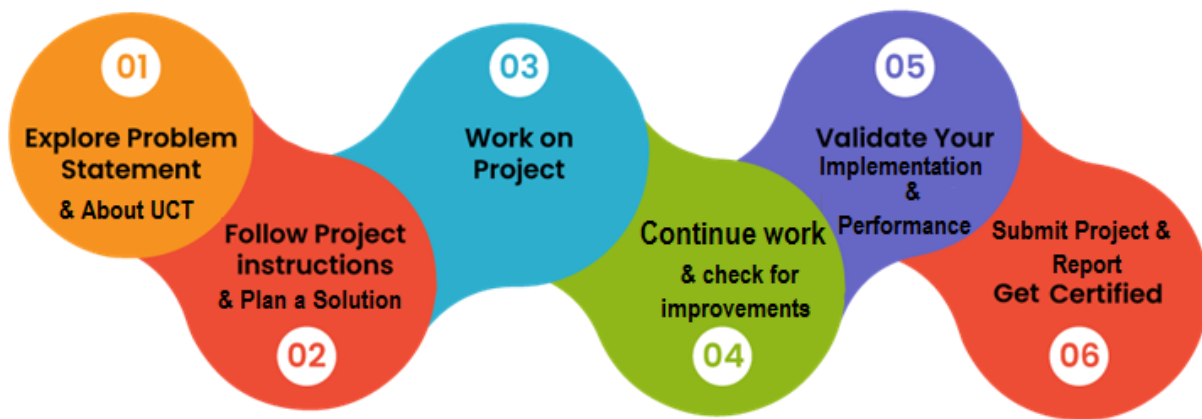
This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This internship was carried out under the Data Science and Machine Learning domain to gain practical experience in applying machine learning techniques to real-world problems. Throughout the internship, I worked on agricultural datasets to understand the complete machine learning pipeline, including data preprocessing, exploratory data analysis (EDA), feature engineering, model development, and performance evaluation.



The primary project, “Prediction of Agriculture Crop Production in India”, focused on building a machine learning model capable of predicting crop production using historical agricultural data. During the internship, I also initiated a second project, “Crop and Weed Detection”, to explore computer vision and object detection concepts using YOLO.

This internship provided valuable exposure to handling real-world datasets, solving practical data science problems, and improving technical as well as analytical skills.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



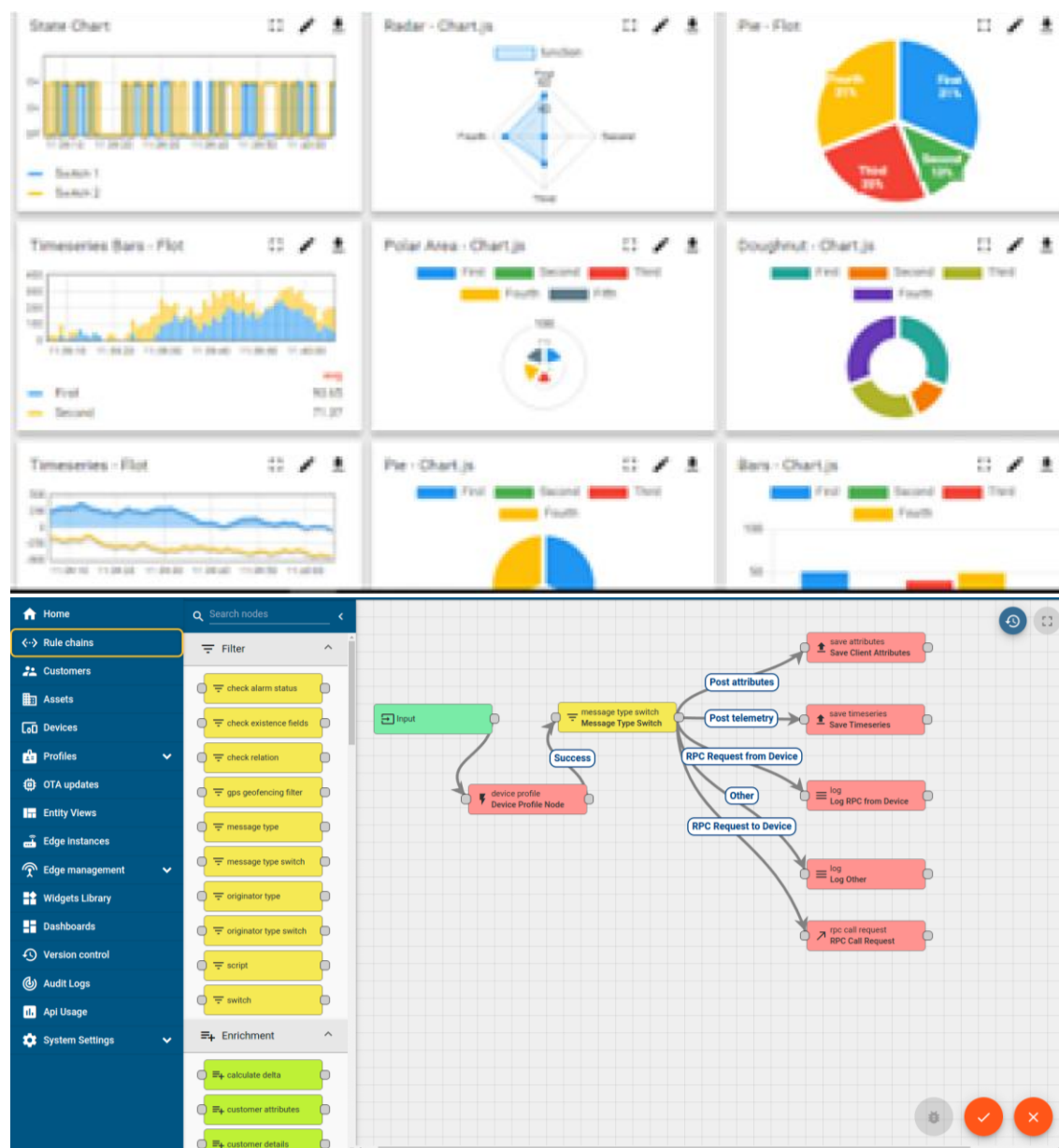
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY **WATCH**

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



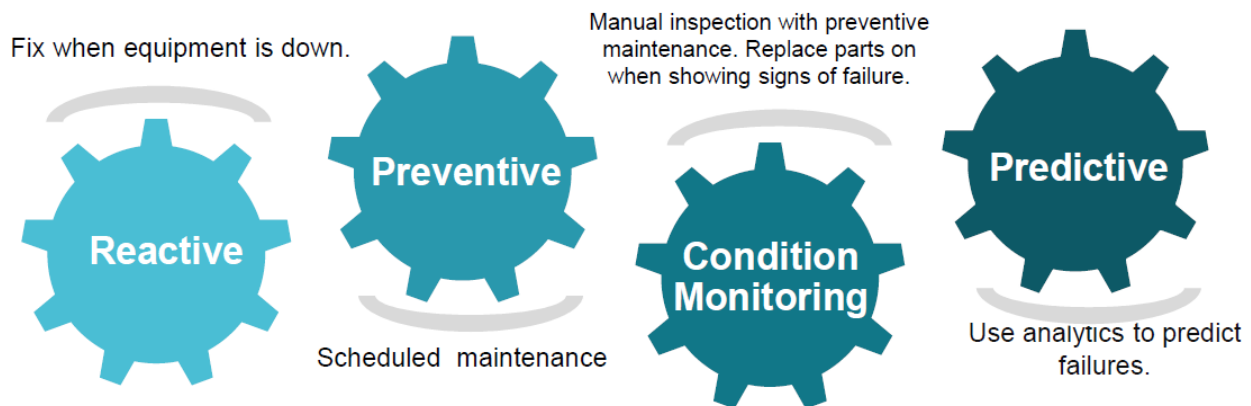


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

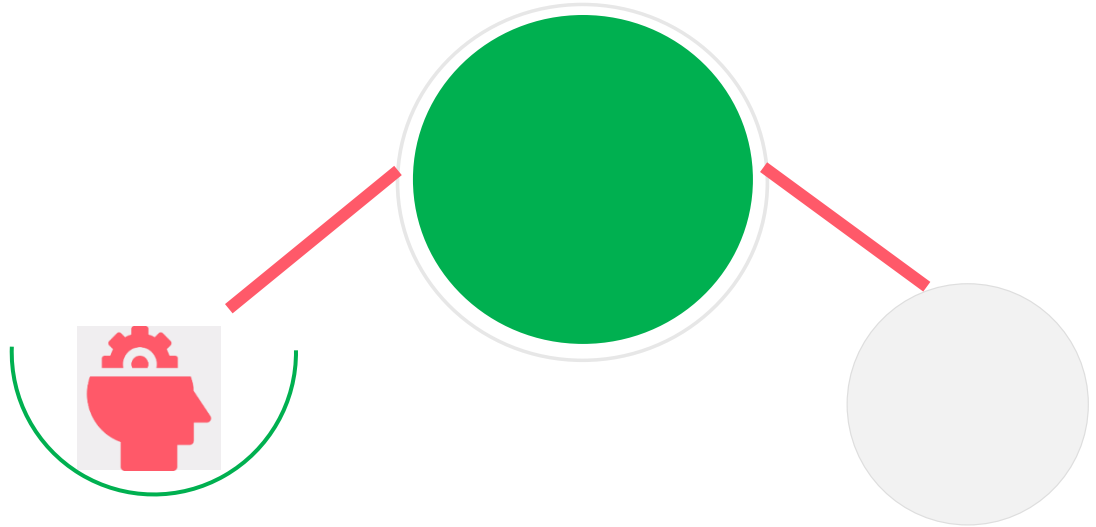
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

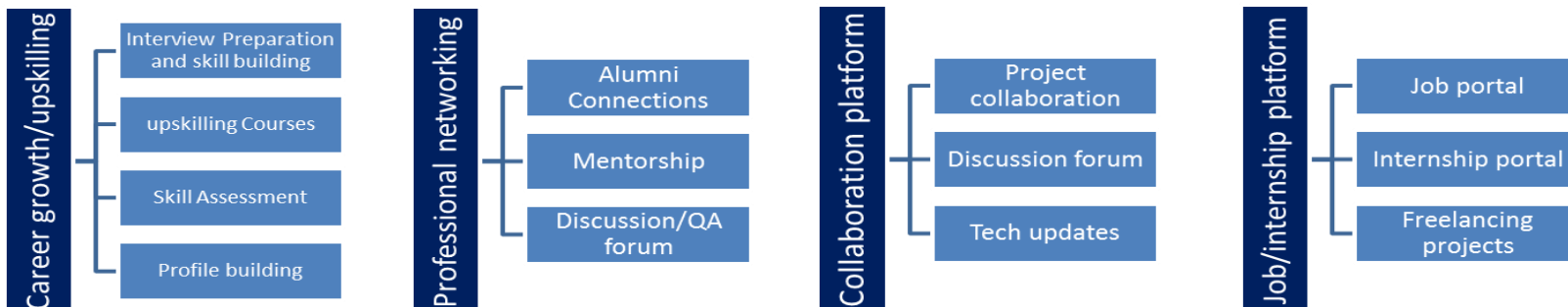
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Documentation
- [2] Kaggle
- [3] Ultralytics YOLO Documentation

2.6 Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
AI	Artificial Intelligence
YOLO	You Only Look Once
CV	Computer Vision

3 Problem Statement

In the assigned problem statement

Crop production is influenced by several factors such as crop type, cultivation area, production cost, yield, seasonal conditions, and geographical location. Accurately estimating crop production is challenging using traditional methods because agricultural data is often large, diverse, and spread across multiple sources.

Agriculture plays a vital role in India's economy and supports the livelihood of a large portion of the population. The objective of this project is to develop a Machine Learning-based system that can predict agricultural crop production in India using historical agricultural data. The project involves collecting and integrating data from multiple datasets, performing data cleaning and preprocessing, conducting exploratory data analysis (EDA), and building a predictive model using suitable machine learning techniques.

In addition to the crop production prediction project, an exploratory study on Crop and Weed Detection was initiated to understand the application of computer vision in agriculture. The aim of this project is to detect crops and weeds from field images using the YOLO object detection framework, supporting precision agriculture through targeted pesticide application and reduced chemical wastage.

Such predictions can assist researchers, agricultural planners, and policymakers in making informed decisions related to crop management and resource planning.

4 Existing and Proposed solution

Crop production estimation is traditionally carried out using historical records, manual surveys, and statistical analysis. As agricultural datasets continue to grow in size and complexity, conventional approaches become less efficient for accurate prediction and decision-making.

Crop Prediction provide useful insights, they are often time-consuming and may not effectively capture the complex relationships between multiple agricultural factors such as crop type, cultivation cost, yield, production area, season, and geographical location. Similarly, weed management in agriculture is commonly performed through manual inspection or blanket pesticide spraying

To address these challenges, a Machine Learning-based approach was adopted for predicting agricultural crop production. Historical agricultural datasets from multiple sources were integrated, cleaned, and preprocessed before performing exploratory data analysis and feature engineering. The trained model achieved satisfactory predictive performance with an R^2 score of approximately 0.75. As an extension of the internship work, a Crop and Weed Detection project was initiated using a dataset of annotated agricultural images in YOLO format. The objective is to develop an object detection system capable of identifying crops and weeds in field images, enabling selective pesticide spraying. This approach has the potential to reduce pesticide wastage, lower cultivation costs, and support precision agriculture practices.

4.1 Code submission (Github link) :

<https://github.com/DhruvChauhan09/upskillcampus/blob/main/PredictionOfAgriculturalCropProductionInIndia.py>

<https://github.com/DhruvChauhan09/upskillcampus/blob/main/CropAndWeedDetectionUsingYOLO.py>

4.2 Report submission (Github link) :

https://github.com/DhruvChauhan09/upskillcampus/blob/main/PredictionOfAgriculturalCropProductionInIndia_Dhruv_USC_UCT.pdf

5 Proposed Design/ Model

Given more details about design flow of the solution.

5.1 High Level Diagram (if applicable)

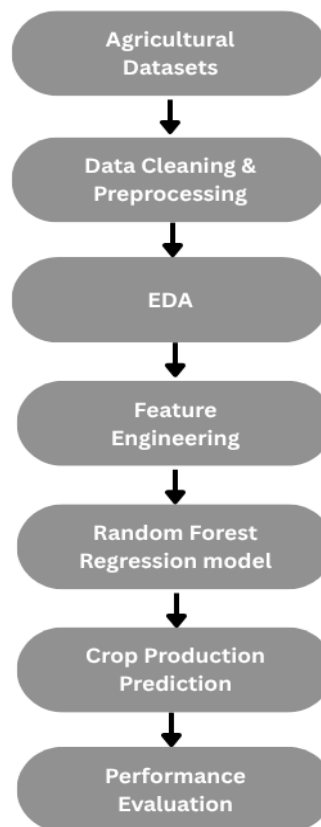


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

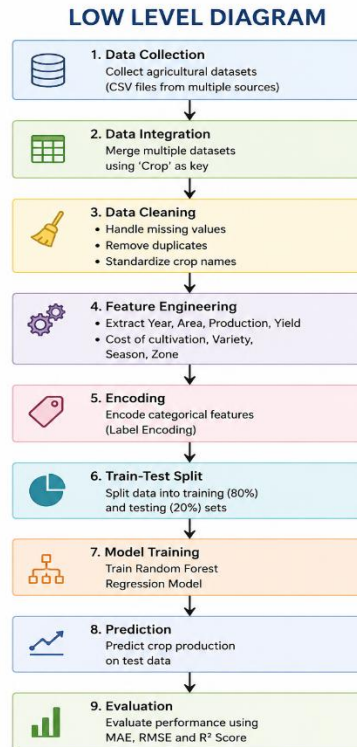


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

- **Programming language** : Python
- **Development environment** : VS Code , Google Colab
- **Libraries** : Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- **Input** : Agricultural CSV datasets

6 Performance Test

6.1 Test Plan/ Test Cases

The developed machine learning model was evaluated to determine its ability to accurately predict agricultural crop production.

The following test plan was followed:

- Load and preprocess the agricultural datasets.
- Merge the datasets based on the crop name.
- Handle missing values and categorical features.
- Split the dataset into training and testing subsets.
- Train the Random Forest Regression model.
- Predict crop production using the testing dataset.
- Evaluate the model using standard regression metrics.

6.2 Test Procedure

The complete workflow followed during model evaluation is summarized below:

- Imported the agricultural datasets into the Python environment.
- Performed data cleaning, preprocessing, and feature engineering.
- Combined multiple datasets to create a unified dataset for model training.
- Encoded categorical variables where required.
- Split the processed dataset into training and testing sets.
- Trained the Random Forest Regression model using the training data.
- Generated predictions for the testing dataset.
- Compared the predicted crop production values with the actual values.
- Calculated model performance using MAE, RMSE, and R^2 Score.

6.3 Performance Outcome

The developed model demonstrated satisfactory predictive performance on the available agricultural dataset.

The evaluation results are summarized below:

Performance Metric	Obtained Value
Mean Absolute Error (MAE)	28.02
Root Mean Square Error (RMSE)	38.59
R ² Score	0.754

The obtained R² score indicates that the model explains approximately 75% of the variation in crop production using the selected input features. The MAE and RMSE values further demonstrate that the prediction errors remain within an acceptable range for the available dataset.

7 My learnings

During this internship, I gained valuable practical experience in the field of Data Science and Machine Learning by working on real-world agricultural datasets. The projects helped me understand the complete machine learning development process, from data collection to model evaluation.

Some of the key learnings from this internship are:

- Improved my Python programming skills for data analysis and machine learning applications.
- Learned how to collect, clean, preprocess, and integrate data from multiple sources.
- Understood the importance of feature engineering and data preprocessing in improving machine learning model performance.
- Learned how to build, train, and evaluate regression models using Scikit-learn, particularly the Random Forest Regression algorithm.
- Strengthened problem-solving and analytical thinking while working with real-world datasets containing missing values and inconsistent information.
- Improved my ability to document project workflows, interpret model results, and present technical work in a structured manner.

Overall, this internship enhanced my technical knowledge, practical implementation skills, and confidence in applying Data Science and Machine Learning techniques to solve real-world problems.

8 Future work scope

The developed crop production prediction model provides a strong foundation for applying Machine Learning in agriculture. However, there are several opportunities to further improve its performance and practical usability.

Future enhancements include:

- Incorporating additional agricultural parameters such as rainfall, temperature, humidity, soil type, fertilizer usage, irrigation facilities, and weather conditions to improve prediction accuracy.
- Expanding the dataset with more recent agricultural records from different states and crop seasons.
- Deploying the trained model as a web application using frameworks such as Streamlit or Flask, allowing users to make crop production predictions through an interactive interface.
- Integrating real-time agricultural data sources to provide more accurate and up-to-date predictions.

In addition to the crop prediction project, the Crop and Weed Detection project can be further developed by training a YOLO-based object detection model using the prepared dataset. The trained model can be used to identify crops and weeds from field images, enabling selective pesticide spraying and supporting precision agriculture.